

DengueRadar: dataset analysis

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Verdict: go. Every part of the senior-first design has usable official data. The national dengue series is clean and covers five outbreaks, live cluster and mosquito-hotspot layers are current, and Census 2020 gives seniors by subzone. The real work is weather: history before 2017 comes from one station, and 2017 to 2024 arrives as roughly 1 GB per year per variable, which must be shrunk to daily totals before it enters Databricks Free Edition.

Weeks of national cases	Outbreak years	Active clusters today (113 cases)	Aedes hotspot areas	Residents aged 65+ (614,380)
574	5	11	135	15.2%

1. Datasets we will use

Core datasets are needed for the MUST scope; Should datasets add depth if time allows. Skipped datasets (stale regional case layers, the 2014 to 2018 duplicate dengue series) are left out.

Dataset (agency)	ID	Coverage	Size	Role in DengueRadar	Tier
Weekly Infectious Disease Bulletin (MOH)	d_ca168b2cb763640d72c4600a68f9909e	2012 to 2022, weekly, national	0.5 MB	Level 1 target: national dengue cases	Core
Dengue Clusters (NEA)	d_dbfabf16158d1b0e1c420627c0819168	Current, updated daily	small	Level 2 signal; daily archive; centre view block list	Core
Areas with High Aedes Population (NEA)	d_5d060d8b7838a15e8906fb22c50dbf51	Current, updated monthly	0.4 MB	Level 2 mosquito signal	Core
Historical Rainfall across Singapore (NEA)	collection 2279 (one per year)	Dec 2016 to 2024, 5-min, all stations	~1 to 1.3 GB/yr	Level 1 lagged rainfall	Core
Historical Air Temperature across Singapore (NEA)	collection 2246	Apr 2016 to 2024, 5-min	~1.2 to 1.4 GB/yr	Level 1 hot-day counts	Core
Realtime Rainfall across Singapore (NEA)	d_6580738cdd7db79374ed3152159fbd69	Live, by station	API	Level 2 local rain (daily job)	Core
Residents by Planning Area/Subzone, Age, Sex, Census 2020 (SingStat)	d_d95ae740c0f8961a0b10435836660ce0	2020, 333 subzone rows	small	Seniors per subzone	Core
Master Plan 2019 Planning Area Boundary (URA)	d_4765db0e87b9c86336792efe8a1f7a66	55 areas	2.1 MB	Island view; spatial joins	Core
Master Plan 2019 Subzone Boundary (URA)	d_8594ae9ff96d0c708bc2af633048edfb	332 subzones	3.2 MB	Senior weighting	Core
Historical Relative Humidity across Singapore (NEA)	collection 2278	May 2016 to 2024, 5-min	~1.2 GB/yr	Humidity feature	Should
Historical Daily Weather Records (NEA)	d_03bb2eb67ad645d0188342fa74ad7066	2009 to Nov 2017, Admiralty only	0.2 MB	Extend training to 2012 to 2016	Should
HDB Property Information (HDB)	d_17f5382f26140b1fdae0ba2ef6239d2f	Every HDB block	1.0 MB	Block age; block list	Should
HDB Existing Building (HDB)	d_16b157c52ed637edd6ba1232e026258d	Block footprints	57 MB	Block locations for centre view	Should
CHAS Clinics (MOH)	d_548c33ea2d99e29ec63a7cc9edcccedc	1,193 clinics	1.6 MB	Nearest subsidised test	Should
Archived NEA clusters (SGCharts; Nature Scientific Data 2020)	external, cited	2013 to 2018, 2020	small	Validate the Level 2 score only	Should

2. How the datasets line up in time

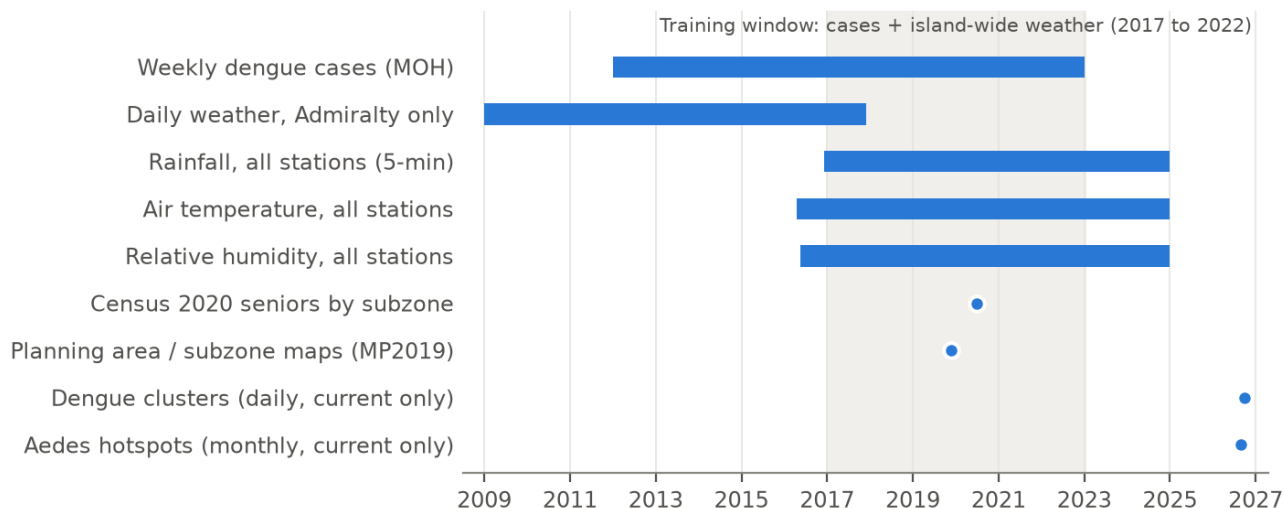


Figure 1. Coverage of each dataset. The shaded band is where weekly cases and island-wide weather overlap.

The overlap between national cases and island-wide weather is **2017 to 2022, about 310 weeks**. That window still contains three outbreaks (2019, 2020, 2022), so the model can be trained on 2017 to 2019 and tested on 2020, then retrained and tested again on 2022. Extending back to 2012 is possible with the single Admiralty station, which covers two more outbreaks (2013, 2014) but gives one location's weather for the whole island. Live layers (clusters, hotspots) exist only as today's snapshot, which is why the daily archive job matters.

3. National dengue cases (Level 1 target)



Figure 2. Weekly lab-confirmed dengue fever cases, Singapore, 2012 to 2022 (MOH). Labels show each outbreak year's peak week.

Year	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022
Cases	4,602	22,101	18,306	11,291	13,153	2,759	3,259	15,910	35,261	5,251	32,130
Peak week	151	838	888	458	635	90	160	661	1,791	194	1,563

Quality: 574 weeks, no duplicate or missing weeks found. The series ends at 2022-W52 on data.gov.sg; recent weeks must come from the Communicable Diseases Agency bulletins. Outbreak years swing from about 2,800 cases (2017) to 35,261 (2020), so the baseline to beat is a recent-average forecast, and error should be judged in outbreak and quiet years separately.

4. Live local layers (Level 2)

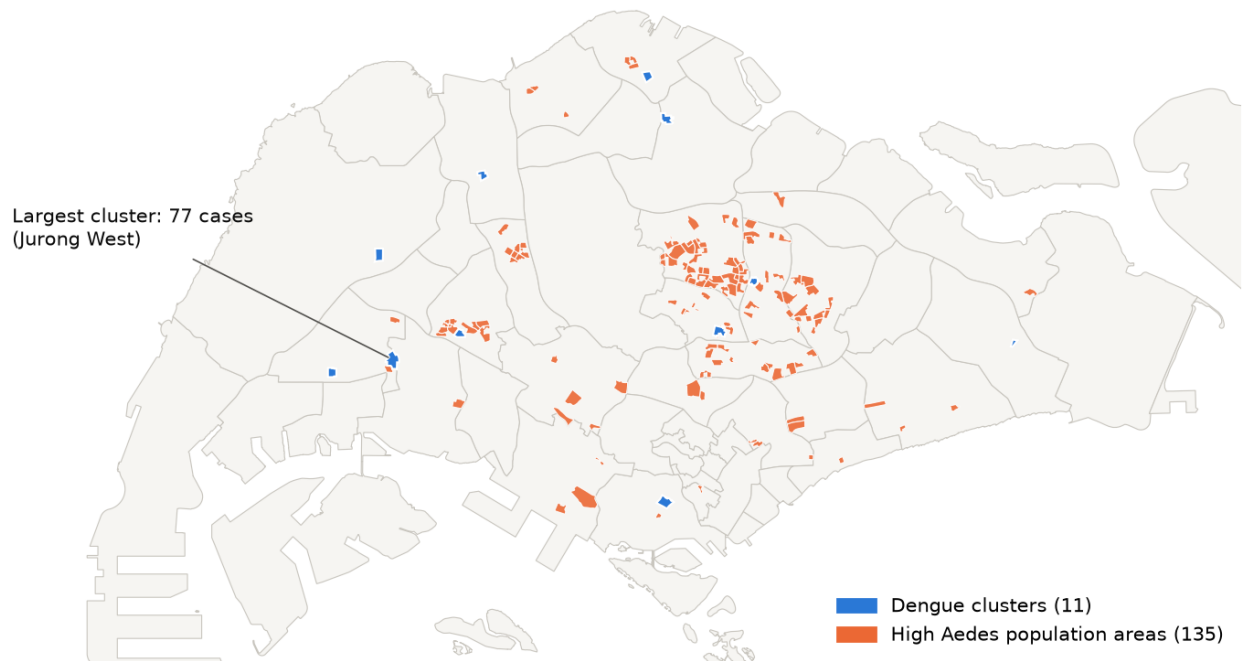


Figure 3. Active dengue clusters and NEA high-Aedes-population areas, snapshot of 27 Sep 2026, over Master Plan 2019 planning areas.

Cluster planning area	Cases	Locality (from NEA)	Share of residents 65+
Jurong West	77	Ho Ching Rd (Blk 117, 119, 120) / Kang Ching Rd (Blk 335) / Tah Ching Rd (Blk 321, 322, 323, 32...)	12.8%
Bishan	8	Bishan St 12 (Blk 109, 110, 111, 115, 116, 118)	18.4%
Sembawang	4	Canberra Rd (Blk 308) / Sembawang Dr / Sembawang Dr (Blk 312) / Sembawang Vista (Blk 318)	8.9%
Jurong West	4	Soon Lee Rd	12.8%
Bukit Merah	4	Kim Tian Rd (Blk 119B, 119D, 127D, 131B)	22.0%
Bukit Batok	4	Bt Batok St 21 (Blk 207, 209, 210)	13.7%
Western Water Catchment	3	Old Choa Chu Kang Rd	1.6%
Tampines	3	Upp Changi Rd East (Changi Ct)	14.7%
Yishun	2	Yishun Ave 5 (Blk 701, 704)	13.0%
Sungei Kadut	2	Mandai Est	20.0%
Serangoon	2	Farleigh Ave / S'goon Gdn Way	19.2%

This is the Level 2 idea in miniature: each cluster already names its blocks, and joining it to Census 2020 shows how senior-heavy the area is (national share 65+: 15.2%). It is low season, so today's layer is small (11 clusters); the 135 Aedes hotspot areas give the score a signal even when clusters are few.

5. Where seniors live (Census 2020)

Highest share 65+ (areas over 20,000 residents)	Share	Most residents 65+	Count
Bukit Merah	22.0%	Bedok	53,370
Ang Mo Kio	21.7%	Tampines	38,260
Queenstown	21.2%	Hougang	36,910
Toa Payoh	21.0%	Ang Mo Kio	35,220
Kallang	21.0%	Jurong West	33,730
Clementi	20.2%	Bukit Merah	33,310
Marine Parade	19.8%	Yishun	28,870
Bedok	19.3%	Toa Payoh	25,630

Share and count point to different places: older estates have the highest share, large towns the most seniors. The Senior Risk Index should use both, and should be computed per subzone rather than planning area so centre staff get block-level lists. Seniors living alone was not found at subzone level; the index starts with share and count only.

6. Weather data: size and plan

Historical rainfall, temperature and humidity are published as one CSV per year of 5-minute station readings: about 1.0 to 1.4 GB each, roughly 30 GB across the three variables for 2017 to 2024. Loading that raw into Databricks Free Edition would waste quota and risk lockouts. Plan:

- Stream each yearly CSV outside Databricks and keep only daily totals (rain), daily max and mean (temperature) and daily mean (humidity) per station. Expected size: a few MB per year.
- Upload the daily tables to bronze; build weekly national and per-station features in silver.
- For 2012 to 2016, test whether Admiralty alone improves the model before relying on it.
- For live scoring, call the realtime rainfall API once a day in the scheduled job and store the result; the demo reads Delta tables, never the live API.

7. Gaps, risks and owners

Gap or risk	Why it matters	Workaround	Owner
Weather files are huge	Raw 5-minute data burns Free Edition quota	Pre-aggregate to daily per station before upload	Jingyi
Weather before 2017 is one station	Limits training to 2017 to 2022 island-wide	Train 2017 to 2022; test Admiralty for 2012 to 2016	Braedon
Cases stop at Dec 2022 on data.gov.sg	Live forecast needs recent weeks	Check Communicable Diseases Agency bulletins 2023 to 2026	Ziqi
Low season: 11 clusters today	Centre view may look empty on Demo Day	Keep daily archive; add Aedes hotspots; record backup demo	Braedon, Ziqi
No seniors-living-alone by subzone	One index factor missing	Use share and count; look for household tables	Ziqi
Eldercare Services list is from 2016	Centre locations out of date	Find a current, public Active Ageing Centre list; cite it	Jingyi
API rate limits (HTTP 429)	Live calls can fail mid-demo	Pause between calls; serve demo from Delta	Braedon
Some dataset IDs return 404	Pipelines can break silently	Record all IDs; re-check the week before Demo Day	Jingyi, Ziqi

Sources

- data.gov.sg datasets listed in section 1 (MOH, NEA, URA, HDB, SingStat), accessed 27 Sep 2026.
- SGCharts Outbreak archive: outbreak.sgcharts.com/data.
- Geographical clusters of dengue outbreak in Singapore during the Covid-19 nationwide lockdown of 2020, Scientific Data (2022).
- Challenge rules on additional public data: daisi.online.